

The prime factorisation of a rectangular product of gcds: a proof of Kaydalov's conjecture on OEIS A092287

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Abstract

For positive integers n, m put $f(n, m) = \prod_{j=1}^n \prod_{k=1}^m \gcd(j, k)$. In March 2019 A. Kaydalov conjectured on OEIS A092287 that $\text{ord}_p f(n, m) = \sum_{i \geq 1} \lfloor n/p^i \rfloor \lfloor m/p^i \rfloor$ for every prime p , reporting verification for $1 \leq m \leq n \leq 200$. We prove it in three lines: $v_p(\gcd(j, k))$ counts the indices i for which p^i divides both j and k , so summing over the rectangle and exchanging the order of summation gives the product of two counting functions at each level i . The square case $m = n$, which is Bala's older conjecture on the same entry (already known to be correct), is the diagonal specialisation.

1 The sequence and the conjecture

OEIS A092287 is

$$a(n) = \prod_{j=1}^n \prod_{k=1}^n \gcd(j, k), \quad a(0), \dots, a(5) = 1, 1, 2, 6, 96, 480.$$

Peter Bala conjectured in April 2007 that $\text{ord}_p a(n) = \sum_{i \geq 1} \lfloor n/p^i \rfloor^2$, and this was confirmed on the entry by C. R. Greathouse IV in April 2013. The entry also carries a second, later comment, which is still open:

Conjecture: For a product over a rectangle, $f(n, m) = \prod_{j=1}^n \prod_{k=1}^m \gcd(j, k)$, a factorisation similar to the one given above for the square case takes place: $\text{ord}_p(f(n, m)) = \lfloor n/p \rfloor \lfloor m/p \rfloor + \lfloor n/p^2 \rfloor \lfloor m/p^2 \rfloor + \dots$. By way of directly computing the values of $f(n, m)$, it can be verified that the conjecture holds at least for all $1 \leq m \leq n \leq 200$. — Andrey Kaydalov, Mar 11 2019

As of the “Last modified” line on the live entry (23 August 2026) this statement is still recorded as an unproven conjecture and no proof appears in the entry's links or programs. This note proves it.

Throughout, p is a prime and $v_p(N)$ denotes the exponent of p in N (written ord_p in the OEIS comment).

2 The theorem

Lemma 1 (layer cake). *For positive integers j, k ,*

$$v_p(\gcd(j, k)) = \min(v_p(j), v_p(k)) = \#\{i \geq 1 : p^i \mid j \text{ and } p^i \mid k\}.$$

Proof. The first equality is the definition of gcd prime by prime. For the second, p^i divides both j and k exactly when $i \leq v_p(j)$ and $i \leq v_p(k)$, i.e. when $1 \leq i \leq \min(v_p(j), v_p(k))$; the number of such i is that minimum. $\square$

Theorem 2. *For all positive integers n, m and every prime p ,*

$$v_p(f(n, m)) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor \left\lfloor \frac{m}{p^i} \right\rfloor, \quad f(n, m) = \prod_{j=1}^n \prod_{k=1}^m \gcd(j, k).$$

The sum is finite: all terms with $p^i > \min(n, m)$ vanish.

Proof. Since v_p is additive on products, Lemma 1 gives

$$v_p(f(n, m)) = \sum_{j=1}^n \sum_{k=1}^m \#\{i \geq 1 : p^i \mid j, p^i \mid k\} = \sum_{i \geq 1} \#\{(j, k) : j \leq n, k \leq m, p^i \mid j, p^i \mid k\},$$

the exchange being a finite rearrangement of a sum of nonnegative terms. The condition on (j, k) splits, so the inner count is $\#\{j \leq n : p^i \mid j\} \cdot \#\{k \leq m : p^i \mid k\} = \lfloor n/p^i \rfloor \lfloor m/p^i \rfloor$. $\square$

Corollary 3. *Taking $m = n$ recovers Bala's square-case formula $\text{ord}_p a(n) = \sum_{i \geq 1} \lfloor n/p^i \rfloor^2$ for A092287.*

Remark (why it is exactly Legendre's formula, squared). The de Polignac–Legendre formula $v_p(n!) = \sum_{i \geq 1} \lfloor n/p^i \rfloor$ is the 1-dimensional version of the same count: $v_p(j) = \#\{i : p^i \mid j\}$, summed over $j \leq n$. Theorem 2 is the 2-dimensional version with a min in place of the single valuation, and the min is precisely what makes the two counting factors separate. The same argument in d dimensions gives $v_p(\prod_{j_1 \leq n_1} \cdots \prod_{j_d \leq n_d} \gcd(j_1, \dots, j_d)) = \sum_{i \geq 1} \prod_{t=1}^d \lfloor n_t/p^i \rfloor$, which covers the companion entry A129454 as well.

Remark (honest assessment). This is a three-line argument and the natural destination is an OEIS comment, not a journal. The only reason it counts as settling something is that the comment was posted in 2019 with a purely numerical justification and has sat unproven since.

3 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(0), \dots, a(15)$ of A092287 directly from the double product and matches the published DATA exactly; and (ii) computes $f(n, m)$ directly from the definition and compares $v_p(f(n, m))$ against $\sum_i \lfloor n/p^i \rfloor \lfloor m/p^i \rfloor$ for every $1 \leq m \leq n \leq 60$ and every prime $p \leq n$, that is 22952 triples (n, m, p) , with 0 mismatches.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A092287>, entry A092287, consulted 23 August 2026.
- [2] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.

A Suggested OEIS comment (plain text, house style)

Kaydalov's rectangle conjecture is true, by the same counting argument that gives Legendre's formula. For a prime p , $v_p(\gcd(j,k)) = \min(v_p(j), v_p(k))$ is the number of $i \geq 1$ with p^i dividing both j and k . Summing over $1 \leq j \leq n$ and $1 \leq k \leq m$ and exchanging the order of summation,

$$\begin{aligned} v_p(f(n,m)) &= \sum_{i \geq 1} \#\{j \leq n : p^i \mid j\} * \#\{k \leq m : p^i \mid k\} \\ &= \sum_{i \geq 1} \text{floor}(n/p^i) * \text{floor}(m/p^i), \end{aligned}$$

since the divisibility conditions on j and k are independent. Setting $m = n$ gives the square case. The same argument in d dimensions gives $v_p(\text{prod over } j_1 \leq n_1, \dots, j_d \leq n_d \text{ of } \gcd(j_1, \dots, j_d)) = \sum_{i \geq 1} \text{floor}(n_1/p^i) * \dots * \text{floor}(n_d/p^i)$; the case $d = 3$ is A129454.